FINAL EXPLANATION: CHEMISTRY GRADE 10

Student Assessment Document

FINAL EXPLANATION: CHEMISTRY GRADE 10 — SUB-STRAND 1.3: THE PERIODIC TABLE	
Student Assessment Document	
Student Name	___________________________________
Class	___________________________________
Date	___________________________________

INSTRUCTIONS FOR STUDENTS

Student Name: _____________________ Class: _____________________ Date: _____________________

NOW IT'S YOUR TURN TO EXPLAIN THE MYSTERY! 🎓

You have completed all 10 lessons in the Periodic Table unit. Think back to Day 1, when you were handed those 20 unlabelled element cards — sodium from the salt on your ugali, chlorine from the Gigiri water-treatment plant, calcium from Maasai cattle bones, carbon from the charcoal glowing under nyama choma. You tried to sort them, and the class ended up with several completely different groupings. Some of you sorted by mass. Some by reactivity. Some by appearance. Nobody could agree — and that confusion was completely valid, because you were missing the invisible key that unlocks everything: electron arrangement.

Now you have that key. This Final Explanation is your opportunity to use it.

Your job is to write a complete, evidence-based scientific explanation that answers the driving question:
"Why do the 118 elements behave so differently, and how can knowing an element's position in the periodic table help us predict what it will do?"

BEFORE YOU WRITE, gather these resources:
✅ Your Summary Table (all 10 lessons)
✅ Your model drawings — from your very first attempt on Day 1 all the way to your Final Model
✅ Data from the element card activities, simulations, and experiments
✅ Your Driving Question Board (DQB) notes — especially questions you added AND questions you answered along the way

AS YOU WRITE, make sure your explanation:
✅ Is written in complete sentences
✅ Uses scientific vocabulary correctly (atomic number, electron configuration, valence electrons, group, period, periodicity, electronegativity, ionisation energy, atomic radius)
✅ Includes specific evidence from at least three different lessons

✅ Shows clear cause-and-effect reasoning (e.g., "Because sodium has one valence electron, it readily loses that electron, which causes…")
✅ Connects electron arrangement to every pattern you describe
✅ References Kenyan contexts from the element cards wherever possible
✅ Reflects how your thinking CHANGED from your initial model to your final model

This is not a test with one right answer — it is your chance to demonstrate deep understanding. Take your time. Be thorough. Be specific. Show everything you know. 🌟

PART 1: The Mystery of the Element Cards — and How Chemists Solved It	
Explain how and why the modern periodic table was developed. In your answer, describe what problem early chemists like Döbereiner and Newlands were trying to solve, what Mendeleev contributed (including his bold decision to leave gaps), and why Moseley's discovery of atomic number was the true turning point. Then explain how the modern periodic table is organised today. Finally, connect this history directly to the Mystery Element Card Challenge: why did different groups in your class sort the cards differently on Day 1, and what was the ONE piece of information that, once understood, would have allowed everyone to sort the cards the same way? **Use evidence from the Data Card Activity and the Historical Development work in Lessons 1 and 2.**	Before the periodic table existed, chemists faced exactly the same confusion you experienced on Day 1 of this unit. They had data about elements — mass, reactivity, appearance — but no agreed system for organising them. Johann Döbereiner noticed that certain elements came in 'triads' of three with similar properties (for example, lithium, sodium, and potassium all react violently with water). John Newlands proposed his 'Law of Octaves,' observing that every eighth element repeated in properties, but this only worked for lighter elements and was widely rejected. The real breakthrough came when Dmitri Mendeleev arranged all 63 known elements in order of increasing atomic mass in 1869 and noticed that properties repeated periodically. His most important and courageous move was leaving deliberate gaps for elements he predicted had not yet been discovered — such as 'eka-aluminium' (later found to be gallium). When those elements were discovered and matched his predictions almost exactly, chemists accepted his table. However, Mendeleev's mass-based arrangement had problems — for example, tellurium (mass ≈ 128) had to be placed before iodine (mass ≈ 127) to keep similar properties together, which broke his own rule. Henry Moseley solved this in 1913 by using X-ray experiments to show that elements should be ordered by atomic NUMBER (the number of protons in the nucleus), not atomic mass. This removed every anomaly and gave us the modern periodic table of 118 elements we use today. The modern periodic table is organised in order of increasing atomic number from left to right and top to bottom. It has 18 vertical columns called groups and 7 horizontal rows called periods. Elements in the same group share similar chemical properties because — as you will explain in Parts 2 and 3 — they have the same number of electrons in their outermost shell. Connecting to the phenomenon: On Day 1, different groups in the class sorted the cards differently because you were using different surface-level clues — mass, colour, reactivity ratings — without knowing the underlying cause of those properties. If everyone had known the atomic numbers on those cards represented the number of protons (and therefore electrons) in each atom, the sorting would have converged on the same answer. The atomic number is the invisible key that the element card photographs hinted at but could not show directly. For example, the sodium card (showing table salt seasoning ugali) and the potassium card both showed high reactivity — and both have atomic numbers that place them in Group 1, meaning they each carry just one electron in their outer shell. You couldn't see that electron arrangement in the photograph, but it was the reason for everything.

PART 2: The Invisible Key — Electron Arrangement	
Explain what electron arrangement (electron	Every atom contains a nucleus (protons and neutrons) surrounded by electrons arranged in energy levels called shells, labelled K, L, M, and N from the inside outward. Each shell has a maximum capacity: K holds up to 2 electrons, L holds up to 8, M holds up to

configuration) is and why it is the fundamental explanation for all periodic table patterns. In your answer, describe electron shells (K, L, M, N), the maximum number of electrons each shell can hold, and how electrons fill shells in order. Show the full electron configurations for at least FIVE elements from the element cards (choose from H, He, C, N, O, Na, Mg, Cl, Ar, Ca). Explain what valence electrons are and why chemists focus on them above all other electrons. Finally, explain how knowing an atom's electron configuration allows you to immediately predict its group and period in the periodic table. **Use evidence from the Build-an-Atom simulation and the electron configuration activities in Lessons 3 and 4.**	18, and N holds up to 32. Electrons always fill the innermost available shell first before moving to the next. This is the rule that connects atomic number directly to electron arrangement. Here are the electron configurations for five elements from the Mystery Element Cards: – Hydrogen (H, atomic number 1): 1 — one electron in the K shell. Configuration: 1. – Carbon (C, atomic number 6): 2, 4 — two electrons in K, four in L. Configuration: 2.4. – Sodium (Na, atomic number 11): 2, 8, 1 — two in K, eight in L, one in M. Configuration: 2.8.1. – Chlorine (Cl, atomic number 17): 2, 8, 7 — two in K, eight in L, seven in M. Configuration: 2.8.7. – Calcium (Ca, atomic number 20): 2, 8, 8, 2 — two in K, eight in L, eight in M, two in N. Configuration: 2.8.8.2. The electrons in the outermost occupied shell are called valence electrons, and they are the only electrons involved in chemical reactions. They are what other atoms 'see' when two atoms meet. This is why chemists focus on valence electrons above all others — the inner shells are full, stable, and chemically inactive. Knowing an element's electron configuration immediately tells you two things about its position in the periodic table: (1) The number of occupied shells equals the element's period number. Sodium has three occupied shells (K, L, M), so it sits in Period 3. Calcium has four occupied shells, so it sits in Period 4. (2) The number of valence electrons equals (or directly determines) the element's group number. Sodium has 1 valence electron → Group 1. Chlorine has 7 valence electrons → Group 17. Calcium has 2 valence electrons → Group 2. Connecting to the phenomenon: In the Build-an-Atom simulation, you could add electrons one at a time and watch the shells fill up. Every time you added an electron that started a new shell, you moved down one period. Every time you increased valence electrons from 1 to 2 to 3 etc., you moved one group to the right. This is why electron arrangement is the invisible key to the element cards — it generates the entire structure of the periodic table automatically, from the inside out.

PART 3: Groups and Families — Why Some Elements Are Almost Identical	
Explain why elements in the same group of the periodic table have similar chemical properties, using valence electrons as your core reasoning. In your answer, focus on at least TWO of the following groups: Group 1 (alkali metals), Group 2 (alkaline earth metals), Group 17 (halogens), and Group 18 (noble gases). For each group you choose, describe the typical properties of the group, explain those properties using valence electron count, and	Elements in the same group have the same number of valence electrons. Because chemical reactions are entirely about gaining, losing, or sharing valence electrons, elements with identical valence electron counts behave in chemically identical — or very similar — ways. This is the answer to one of the most puzzling questions from Day 1: how can two elements separated by many others in atomic number act almost the same? They act the same because their outermost electron shell looks the same. Group 1 — Alkali Metals (Li, Na, K…): Every alkali metal has exactly 1 valence electron. That single outer electron is loosely held and very far from the nucleus, making it extremely easy to lose. When an alkali metal loses its 1 valence electron, it becomes a stable positive ion (e.g., Na^+, K^+) with a full outer shell — and elements always 'want' a full outer shell (the octet rule). This makes all alkali metals highly reactive, soft, shiny, and low in density. From the element cards, sodium (the card showing table salt used to season ugali) is a perfect example: Na reacts rapidly with water and air, and in salt form (NaCl) it has already lost that 1 valence electron to chlorine. Potassium behaves nearly identically — because it also has 1 valence electron — but is even more reactive because its valence electron is in a higher, more distant shell and thus even easier to remove. Group 17 — Halogens (F, Cl, Br…): Every halogen has exactly 7 valence electrons — just one short of a full shell of 8. This makes halogens extremely eager to GAIN 1 electron. From the element cards, chlorine (the card showing water treatment at Nairobi's

give at least one Kenyan-context example from the element cards. Then explain what happens when elements from Group 1 meet elements from Group 17 — why do they react so readily with each other? **Use evidence from the reactivity experiments and simulations in Lessons 5 and 6.**	Gigiri plant) is an excellent example: Cl readily gains 1 electron to become Cl^-, a stable chloride ion with a full outer shell. This powerful electron-pulling ability (high electronegativity) makes chlorine an effective disinfectant, because it attacks and destroys microorganism cell membranes by reacting with their chemical bonds. Group 1 meets Group 17 — a perfect reaction: When sodium (1 valence electron to give away) meets chlorine (desperate for 1 electron), the reaction is immediate and energetically very favourable. Sodium loses its electron; chlorine gains it. Both achieve full outer shells. The product is sodium chloride (NaCl) — table salt — the very substance photographed on the sodium element card. This is not a coincidence; it is the periodic table predicting chemistry. Group 18 — Noble Gases (He, Ne, Ar…): Every noble gas already has a completely full outer shell (He has 2 electrons filling its K shell; Ne and Ar have 8 electrons filling their outer shells). With nothing to gain and nothing to lose, noble gases almost never react. This explains why argon is used to create inert atmospheres in industrial welding and why helium balloons do not combust. Connecting to the phenomenon: In the reactivity experiments, you observed that lithium, sodium, and potassium all reacted with water — but with increasing violence — while the noble gas argon did nothing at all. The electron configuration explains both facts perfectly. The element cards were showing you the OUTCOMES of valence electron behaviour; now you can explain the CAUSE.

PART 4: Trends Across Periods — Why Elements NEXT to Each Other Can Be Completely Different	
Explain the trends that occur as you move ACROSS a period (left to right) in the periodic table. In your answer, describe and explain the trends in: (1) atomic radius, (2) ionisation energy, and (3) metallic vs. non-metallic character. Use Period 3 (Na → Cl) as your main example, and reference specific elements from the element cards wherever possible. Then answer the second part of the Day 1 puzzle: why can elements sitting RIGHT NEXT TO EACH OTHER in the periodic table behave completely differently? **Use evidence from the periodic trends simulation and data analysis activities in Lessons 7 and 8.**	While elements in the same GROUP are similar, elements in the same PERIOD can be dramatically different — and this was the second great puzzle of the Mystery Element Card Challenge. The answer lies in understanding what changes as you add protons one at a time across a period. Across Period 3 — from sodium (Na, atomic number 11) to chlorine (Cl, atomic number 17) — the number of protons in the nucleus increases from 11 to 17. Crucially, all Period 3 elements are adding electrons into the SAME M shell. This means the nuclear charge (positive pull) increases, but the shielding from inner shells stays constant (always 2 electrons in K and 8 in L). The result is that the nucleus pulls the outer electrons in more and more tightly as you move right. Trend 1 — Atomic Radius DECREASES across a period: As nuclear charge increases from Na to Cl, the stronger pull drags the electron cloud inward. Sodium atoms are significantly larger than chlorine atoms, even though chlorine has more electrons. Larger atoms (left of period) lose electrons more easily; smaller atoms (right of period) hold electrons more tightly. Trend 2 — Ionisation Energy INCREASES across a period: Ionisation energy is the energy required to remove one electron from an atom. Because the nucleus pulls harder on electrons as you move right, it takes more and more energy to remove an electron. Sodium (1 valence electron, large atom, low nuclear pull on outer electron) has a very low first ionisation energy — it gives up its electron almost eagerly. Chlorine (7 valence electrons, small atom, high nuclear pull) has a very high ionisation energy — it will not give up any electrons at all; instead it pulls electrons FROM other atoms. Trend 3 — Metallic Character DECREASES across a period: On the left of Period 3, sodium and magnesium are shiny, conducting metals that lose electrons easily — classic metallic behaviour. In the middle, silicon is a metalloid. On the right, phosphorus, sulfur, and chlorine are non-metals that gain electrons rather than lose them. This is why the left side of the periodic table is the 'metal zone' and the right side is the 'non-metal zone.'

	Why elements next to each other can be completely different: Sodium (Na, Group 1) and magnesium (Mg, Group 2) sit right next to each other in Period 3. Yet sodium reacts explosively with water while magnesium reacts only slowly. The reason: sodium has 1 valence electron (extremely easy to lose) while magnesium has 2 valence electrons (the second one is harder to remove because the nuclear charge is higher). One extra proton and one extra valence electron create a measurable difference in reactivity. Moving further right, from magnesium to aluminium to silicon to phosphorus, the character shifts from metal to metalloid to non-metal. By the time you reach chlorine — just six positions to the right of sodium — you have moved from a soft, explosive metal to a toxic, electron-hungry gas. The periodic table is not a gentle gradient; each step across a period is a meaningful chemical change. Connecting to the phenomenon: On Day 1, some students sorted element cards by appearance and were confused when sodium (shiny metal) and chlorine (yellow-green gas) ended up in the same period. Now you understand: same period means same number of electron shells, but completely different valence electron counts and nuclear charges — and that produces completely different chemistry.

PART 5: Your Complete Answer to the Driving Question — and Your Model Journey	
Now bring everything together. Write a complete, connected explanation that directly answers the driving question: 'Why do the 118 elements behave so differently, and how can knowing an element's position in the periodic table help us predict what it will do?' Your answer must: (1) explain why elements behave differently using atomic structure and electron arrangement; (2) explain how position in the periodic table (group AND period) allows prediction of properties; (3) reference at least THREE specific elements from the original Mystery Element Cards with their Kenyan contexts; (4) describe how your thinking changed from your initial model on Day 1 to your Final Model today — what did your first model get wrong or leave out, and what does your Final Model include that it did not?	The 118 elements behave so differently from one another because each element has a unique number of protons in its nucleus — its atomic number — and this determines how many electrons the atom has and how those electrons are arranged in shells. It is the arrangement of electrons, especially the number of valence electrons in the outermost shell, that directly controls how an element behaves chemically. Two elements with the same number of valence electrons will behave similarly, regardless of how far apart they are on the table. Two elements with different numbers of valence electrons will behave differently, even if they sit right next to each other. Knowing an element's position in the periodic table gives you two immediate, powerful predictions: (1) Its PERIOD number tells you how many electron shells the atom has, which tells you roughly how large the atom is and how tightly the nucleus holds onto the outer electrons. Elements in higher periods have more shells, larger radii, and lower ionisation energies. (2) Its GROUP number tells you how many valence electrons the atom has, which tells you whether it tends to lose electrons (left side, low group numbers), gain electrons (right side, high group numbers), or neither (Group 18, noble gases). From these two facts alone, you can predict metallic character, reactivity, ion formation, and the type of bonds the element will form. Three elements from the Mystery Element Cards illustrate this perfectly: – Sodium (Na, the ugali salt card) is in Period 3, Group 1: 3 electron shells, 1 valence electron. Prediction: very reactive metal, loses 1 electron easily, forms Na^+ ions, reacts with non-metals like chlorine. Confirmed: NaCl is table salt. – Chlorine (Cl, the Gigiri water treatment card) is in Period 3, Group 17: 3 electron shells, 7 valence electrons. Prediction: reactive non-metal, gains 1 electron easily, forms Cl^- ions, reacts with metals. Confirmed: used to disinfect water because it attacks chemical bonds in microorganisms. – Calcium (Ca, the Maasai cattle bones card) is in Period 4, Group 2: 4 electron shells, 2 valence electrons. Prediction: reactive metal, loses 2 electrons to form Ca^{2+}, bonds with non-metals. Confirmed: calcium in bones exists as calcium phosphate, where Ca^{2+} ions bond with phosphate groups. In every case, the element card photograph showed you the real-world outcome, and the periodic table position explains the chemical cause.

This is your most important section. Write it carefully and completely.	My Model Journey: On Day 1, my initial model was incomplete and surface-level. I sorted the element cards based on [describe your own approach — e.g., mass, appearance, or reactivity rating]. My model had no explanation for WHY certain elements were similar; it just observed that they were. It could not predict anything — it could only describe what the cards already told me. My drawings of atoms, if I drew any, probably showed a simple blob with a nucleus and some electrons scattered around it, with no shells and no connection to the periodic table layout. By the end of Lesson 4, my model improved to include electron shells and configurations. By the end of Lesson 7, it included group and period trends. My Final Model today shows: (1) atoms with correctly filled electron shells (K, L, M, N); (2) valence electrons specifically labelled; (3) arrows showing the trend in atomic radius and ionisation energy across a period; (4) arrows showing increasing reactivity down Group 1 and increasing electronegativity across to Group 17; (5) labels connecting each element back to its Kenyan context card. Most importantly, my Final Model includes the CAUSE (electron arrangement) linked by arrows to the EFFECT (observable properties like reactivity, state, metallic character) — which my initial model did not have at all. The Mystery Element Card Challenge is now solved: the invisible internal structure — the number and arrangement of electrons — is the one factor that explains every pattern on those cards. Mendeleev guessed at this pattern from the outside, using mass and properties. Moseley found the key with atomic number. Quantum chemistry revealed the electron shells. And now, holding any one of those 118 element cards, you can look up its position in the periodic table and predict what it will do — not by memorising facts, but by understanding the electron arrangement that drives all chemistry.

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Criterion	**Excellent (4)**	**Proficient (3)**	**Developing (1–2)**
Scientific Accuracy of Electron Configuration & Atomic Structure	All electron configurations are written correctly. The student accurately describes shells (K, L, M, N), maximum capacities, and filling order. Valence electrons are correctly identified for all elements discussed. The link between atomic number, electron arrangement, and periodic table position is explained precisely and without error.	Most electron configurations are correct with at most one minor error. The student correctly identifies valence electrons for the majority of elements discussed and makes a clear connection between electron arrangement and periodic table position.	Several electron configuration errors are present. The student shows partial understanding of shells and valence electrons but confuses or omits the connection between electron arrangement and an element's position in the table.
Explanation of Periodic Trends (Groups and Periods)	The student accurately explains trends in atomic radius, ionisation energy, and metallic/non-metallic character both across periods and down groups, with correct cause-and-effect reasoning (e.g., increasing nuclear charge across a period with constant shielding). At least two groups are discussed in depth with correct property descriptions linked to valence electron count.	The student correctly explains at least two periodic trends and discusses at least two groups with mostly accurate reasoning. Cause-and-effect language is used but may be incomplete in one area.	The student identifies that trends exist but provides limited or partially incorrect explanations. Valence electrons may be mentioned without being used to explain the trend. Group properties may be listed without explanation.
Use of Evidence from Lessons	The student references specific evidence	The student references evidence from at	The student includes some reference to

and Kenyan Contexts	from at least three different lessons (e.g., Build-an-Atom simulation, reactivity experiments, periodic trends data). At least three Kenyan-context elements from the Mystery Element Cards are cited by name with their context (e.g., sodium/ugali salt, chlorine/Gigiri water treatment, calcium/Maasai cattle bones), and the context is used meaningfully to illustrate a chemical concept.	least two lessons and includes at least two Kenyan-context elements from the element cards. The evidence is connected to the explanation but may not always be used to directly support a specific claim.	class activities or Kenyan contexts but the evidence is vague, minimal (only one context), or not clearly connected to the chemical explanation being made.
Direct Answer to the Driving Question with Reasoning	Part 5 provides a complete, logically structured answer to the driving question. The student explains BOTH why elements behave differently (electron arrangement / atomic number) AND how position predicts behaviour (group = valence electrons; period = number of shells). Cause-and-effect reasoning is explicit and consistent throughout. The answer stands alone as a coherent scientific explanation.	The student addresses both parts of the driving question with mostly correct reasoning. The explanation is logical and uses scientific vocabulary correctly, though some cause-and-effect links may be implicit rather than explicitly stated.	The student attempts to answer the driving question but addresses only one part (e.g., only explains why elements differ, or only describes how the table is organised without linking this to prediction). Reasoning is present but incomplete or partially incorrect.
Model Journey — Reflection on Conceptual Change	The student provides a specific and honest description of their initial model's limitations (what it got wrong or left out) and a detailed description of what their Final Model includes that the initial model did not. The reflection demonstrates genuine conceptual change — the student can articulate WHAT changed in their thinking and WHY, linking changes to specific lessons or evidence encountered during the unit.	The student describes both their initial and final models and notes at least two specific differences between them. The reflection shows awareness of conceptual growth, though the explanation of what caused the change may be general rather than tied to specific lessons or evidence.	The student describes their final model but gives only a vague or very brief account of how it differs from their initial model. The reflection may state that 'I learned more' without identifying specific conceptual changes or the evidence that drove those changes.